

Fernando Lucatelli Nunes

Assistant Professor, University of Coimbra

arXiv preprints: https://arxiv.org/a/lucatellinunes_f_1.html

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Mathematics Genealogy, Google Scholar, Scopus ID 57188838707

Assistant Professor of Mathematics at the University of Coimbra, holding a PhD in Mathematics (2018) and a forthcoming second PhD in Computer Science (2026). My research develops structural mathematics at a high level of abstraction, addressing intrinsic problems in pure mathematics while establishing foundations for emerging programming paradigms. In pure mathematics, my work spans generalised categorical structures, two- and three-dimensional universal algebra, Grothendieck descent theory, and categorical Galois theory, with deep connections to algebraic topology, geometry, and logic. In computer science, I apply these structural methods to programming language semantics, correctness-by-construction program transformations, and the mathematical foundations of differential, probabilistic, hybrid, and quantum programming. My aim is to lead a research programme that uses categorical and algebraic ideas to solve substantial mathematical problems and to develop conceptual frameworks for reliable, transparent, and mathematically robust computational systems.

ACADEMIC APPOINTMENTS

2026 – present	Assistant Professor , Department of Mathematics, University of Coimbra, Portugal.
2025 – 2026	Research Fellow , School of Computer Science, University of Birmingham, United Kingdom.
2025	Postdoctoral Fellow , Department of Information and Computing Sciences, Utrecht University, The Netherlands.
2020 – 2025	Doctoral Researcher , Department of Information and Computing Sciences, Utrecht University, The Netherlands.
2023	Postdoctoral Fellow , Department of Mathematics and Statistics, York University, Canada.
2023	Fields Research Fellow , Fields Institute for Research in Mathematical Sciences, Canada.
2022	Oberwolfach Leibniz Fellow , Mathematisches Forschungsinstitut Oberwolfach, Germany.
2018 – 2020	Research Fellow , Department of Mathematics, University of Coimbra, Portugal.
2018 – 2019	Invited Assistant Professor , Department of Mathematics, University of Coimbra, Portugal.
2018	Postdoctoral Fellow , University of Coimbra, Portugal.
2012 – 2018	Doctoral Researcher , Department of Mathematics, University of Coimbra, Portugal.

VISITING ACADEMIC APPOINTMENTS

2026	Visiting Researcher , Utrecht University.
2025	Visiting Researcher , IMPA, Rio de Janeiro, Brazil.
2018, 2023, 2024, 2025	Visiting Researcher , UCLouvain, Belgium.
2019, 2025	Visiting Researcher , University of Cape Town, South Africa.
2019, 2025	Visiting Researcher , Stellenbosch University, South Africa.

RESEARCH GROUPS AND AFFILIATIONS

2026 – present	Integrated Researcher , Centre for Mathematics, University of Coimbra (CMUC), Portugal.
2025 – 2026	Research Fellow , Theory of Computation, School of Computer Science, University of Birmingham, United Kingdom.
2020 – 2026	Researcher in Programming Language Semantics , Software Technology Group, Utrecht University, The Netherlands.
2020 – 2024	Research Collaborator , Centre for Mathematics, University of Coimbra (CMUC), Portugal.
2018 – 2020	Postdoctoral Researcher, and Research Fellow , Centre for Mathematics, University of Coimbra (CMUC), Portugal.
2012 – 2018	Researcher in Pure Mathematics , Centre for Mathematics, University of Coimbra (CMUC), Portugal.
2012 – 2018	Researcher in Category Theory , Coimbra School of Category Theory, Portugal.

RESEARCH PROFILE

In pure mathematics, my research develops the theory and applications of generalised categorical structures in low and higher dimensions, primarily through two- and three-dimensional universal algebra. This work addresses intrinsic questions in categorical algebra, Grothendieck descent theory, categorical Galois theory, and related structural fields, while maintaining deep connections with proof theory, the geometry of manifolds, algebraic topology, homotopy theory, and logic.

In computer science, my research focuses on programming language semantics, formal methods, and software technology, with particular emphasis on correctness-by-construction methodologies and principled program transformations. I apply categorical and structural methods to emerging programming paradigms, including differential, probabilistic, hybrid, and quantum programming.

Across these areas, my aim is to lead a research programme that brings deep categorical and algebraic ideas to bear on substantial problems in pure mathematics and on the foundations of modern computation. A central objective of this programme is to train researchers who can move confidently between intrinsically motivated mathematics and foundational as well as practical challenges in computer science.

Education

PhD IN MATHEMATICS (AWARDED 2018)

10/2012 – 01/2018

University of Coimbra, PhD in Mathematics

- Thesis: *Pseudomonads and Descent* (awarded *summa cum laude*).
- Supervisor: Prof. Maria Manuel Clementino.
- PhD awarded in Jan/2018
- Examination Committee: Prof. Martin Hyland, Prof. Marco Mackaay, Prof. Lurdes Sousa, and Prof. Peter Beier Gothen.
- Keywords: Biadjoint triangle theorems; freely generated categorical structures; theories of presentation and coherence for low- and higher-dimensional categorical structures; Grothendieck descent theory; categorical Galois theories; low- and higher-dimensional weak limits.

PhD IN COMPUTER SCIENCE (to be awarded in September, 2026)

11/2020 – 11/2025

Utrecht University, PhD in Computer Science

- Thesis: *Freely Generated Categorical Structures and Automatic Differentiation*.
- Supervisors: Prof. Gabriele Keller and Dr. Matthijs Vákár.
- PhD to be awarded Sep/2026
- Examination Committee: Prof. Ieke Moerdijk, Prof. Bart Jacobs, Prof. Sam Staton, Prof. Robin Cockett, and Prof. Patricia Johann.
- Keywords: Denotational semantics of programming languages; automatic differentiation; principled program transformations; recursion and iteration; categorical semantics; logical relations; type systems and dependent types; Grothendieck constructions; extensive categories.

SELECTED PUBLICATIONS

- Lucatelli Nunes, F., Vákár, M.: **Monoidal closure of Grothendieck constructions via Σ -tractable monoidal structures and Dialectica formulas.** Theory Appl. Categ. 44, 1153–1217. 2025. (65 pages).
- Lucatelli Nunes, F., Prezado, R., Vákár, M.: **Free extensivity via distributivity.** Portugaliae Mathematica, 82, no. 1/2, pp. 177–204. 2025. (28 pages).
- Prezado, R., Lucatelli Nunes, F.: **Generalized multicategories: change-of-base, embedding, and descent.** Appl. Categ. Struct. Volume 32, 35. 2024. (89 pages).
- Clementino, M.M., Lucatelli Nunes, F.: **Lax comma 2-categories and admissible 2-functors.** Theory Appl. Categ. 40, 180–226. 2024. (47 pages).
- Clementino, M.M., Lucatelli Nunes, F., Prezado, R.: **Lax comma categories: cartesian closedness, extensivity, topologicity, and descent.** Theory Appl. Categ. 41, 516–530. 2024. (15 pages).
- Lucatelli Nunes, F., Vákár, M.: **Automatic differentiation for ML-family languages: Correctness via logical relations.** Math. Struct. Comput. Sci. 34, No. 8, 747–806. 2024. (60 pages).
- Prezado, R., Lucatelli Nunes, F.: **Descent for internal multicategory functors.** Appl. Categ. Struct. 31, 11. 2023. (18 pages).
- Clementino, M.M., Lucatelli Nunes, F.: **Lax comma categories of ordered sets.** Quaest. Math. 46, Part Suppl. 1, 145–159. 2023. (14 pages).
- Lucatelli Nunes, F., Vákár, M.: **CHAD for Expressive Total Languages.** Math. Struct. Comput. Sci. 33, No. 4–5, 311–426. 2023. (116 pages).
- Lucatelli Nunes, F., Prezado, R., Sousa, L.: **Cauchy Completeness, Lax Epimorphisms, and Effective Descent for Split Fibrations.** Bull. Belg. Math. Soc. 30, No. 1, 130–139. 2023. (10 pages).
- Lucatelli Nunes, F.: **Semantic factorization and descent.** Appl. Categ. Struct. 30, No. 6, 1393–1433. 2022. (41 pages).
- Lucatelli Nunes, F., Sousa, L.: **On lax epimorphisms and the associated factorization.** J. Pure Appl. Algebra 226, No. 12, 27 p. 2022. (27 pages).
- Lucatelli Nunes, F.: **Descent data and absolute Kan extensions.** Theory Appl. Categ. 37, 530–561. 2021. (32 pages).
- Lucatelli Nunes, F.: **Pseudoalgebras and non-canonical isomorphisms.** Appl. Categ. Struct. 27, No. 1, 55–63. 2019. (9 pages).
- Lucatelli Nunes, F.: **On lifting of biadjoints and lax algebras.** Categ. Gen. Algebr. Struct. Appl. 9, No. 1, 29–58. 2018. (30 pages).
- Lucatelli Nunes, F.: **Pseudo-Kan Extensions and Descent Theory.** Theory Appl. Categ. 33, 390–444. 2018. (35 pages).
- Lucatelli Nunes, F.: **On biadjoint triangles.** Theory Appl. Categ. 31, 217–256. 2016. (40 pages).

ACCEPTED FOR PUBLICATION

- Lucatelli Nunes, F., Plotkin, G., Vákár, M.: **Unraveling the iterative CHAD.** arXiv:2505.15002. 2025. (57 pages). (Mathematical Structures in Computer Science.)
- Lucatelli Nunes, F., Vákár, M.: **Free doubly-infinitary distributive categories are cartesian closed.** arXiv:2403.10447. 2024. (16 pages). (Applied Categ. Struct.)

SELECTED MANUSCRIPTS IN PREPARATION

Lucatelli Nunes, F., and Tholen, W.: **Spanning the tale of Monads and Descent**. Manuscript in preparation

Lucatelli Nunes, F., and Clementino, M.: **On lax comma 2-categories**. Manuscript in preparation

Lucatelli Nunes, F., Plotkin, G., and Vákár, M.: **Backprop to the Future: CHAD for Recursion**. Manuscript in preparation

Lucatelli Nunes, F., Goncharov, S., Vákár, M., and Van der Linden, T.: **Symmetric Infinitary Monoidal Categories**. Manuscript in preparation

Lucatelli Nunes, F., and Levy, P.B.: **In-place strictification**. Manuscript in preparation

PREPRINTS (UNDER REVIEW)

Lucatelli Nunes, F., Prezado, R.: **Functors Preserving Effective Descent Morphisms**. arXiv:2410.22876. 2024. (*17 pages*).

Lucatelli Nunes, F.: **Freely generated n-categories, coinserter and presentations of low dimensional categories**. arXiv:1704.04474. 2017. (*56 pages*).

OTHER PUBLICATIONS

Lucatelli Nunes, F., Plotkin, G., Vákár, M.: **Homomorphic Reverse Differentiation of Iteration**. LAFI 2024 (Tenth Workshop on Languages for Inference at POPL, 2024). 2024. (*2 pages*).

Lucatelli Nunes, F., Vákár, M.: **Logical Relations for Partial Features and Automatic Differentiation Correctness**. arXiv:2210.08530. 2022. and Oberwolfach Preprints, OWP-2023-09. 2023. (*21 pages*).

Lucatelli Nunes, F.: **Freely generated categorical structures and automatic differentiation**. Computer Science PhD Thesis, Utrecht University. 2025. (*380 pages*).

Lucatelli Nunes, F.: **Pseudomonads and Descent**. Mathematics PhD Thesis, University of Coimbra. 2017. (*209 pages*).

Lucatelli Nunes, F.: **Espaços não reversíveis**. (Non-reversible spaces). Revista Matemática Universitária, v. 48, n. 49, p. 22-26. 2012. (*5 pages*).

UNDERGRADUATE AND GRADUATE MONOGRAPHS

Lucatelli Nunes, F.: **Basic Concepts of Topology for Topological Dynamics (in Portuguese)**. 2009. (*116 pages*).

Lucatelli Nunes, F.: **Basic Topological Dynamics and Basic Applications to Number Theory (in Portuguese)**. 2009. (*57 pages*).

Grants, Fellowships, and Funded Projects

SELECTED RESEARCH FELLOWSHIPS (PI)

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| 2022 | Oberwolfach Leibniz Fellowship , Grothendieck Descent Theory, Galois Theory, and Presentations of Categorical Structures <ul style="list-style-type: none">• PI: Fernando Lucatelli Nunes.• Participants: Maria Manuel Clementino, Rui Prezado, Lurdes Sousa, Matthijs Vákár. |
| 2023 | Fields Research Fellowship , Grothendieck Descent Theory, Fibrations, and Factorizations <ul style="list-style-type: none">• PI: Fernando Lucatelli Nunes.• Participants: Walter Tholen. |

SELECTED PARTICIPATION IN RESEARCH PROJECTS AND GRANTS

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| 2025 – present | ERC Starting Grant FoRECAST , Formalised Reasoning about Expectations: Composable, Automated, Speedy, Trustworthy <ul style="list-style-type: none">• PI/Grant holder: Matthijs Vákár.• My role: Member of the formal methods team, contributing to the design of principled algorithms grounded in categorical structures, and involved in the (co-)supervision of PhD students. |
| 2025 – 2026 | DFG Project HOMBRe , Higher-Order Monad-based Programming and Reasoning <ul style="list-style-type: none">• PI/Grant holder: Sergey Goncharov.• My role: Postdoctoral Fellow, contributing to the core research themes of the project. |
| 2021 – 2024 | NWO Veni Grant (VI.Veni.202.124) , Correct Expressive Differential Programming: Laying the Mathematical Foundations for Tomorrow’s Machine Learning <ul style="list-style-type: none">• PI/Grant holder: Matthijs Vákár.• My role: Led the work on Iterative CHAD, contributing to the formal methods component of the project. Additionally, established the correctness of dual-numbers-based (forward- and reverse-mode) automatic differentiation for languages with recursive types. |
| 2020 – 2022 | Marie Skłodowska-Curie Grant , Principled AD Program Transformations <ul style="list-style-type: none">• PI/Grant holder: Matthijs Vákár.• My role: Developed the correct extension of reverse-mode automatic differentiation (CHAD) program transformations to expressive total languages. |
| 2011 – 2011 | Project: Nonlinear Analysis and Applications (CNPq) , Group Project (University of Brasília), funded by CNPq <ul style="list-style-type: none">• Project title: Nonlinear Analysis and Applications.• My role: Undergraduate student; led the subproject “Topological Dynamics and Applications in Number Theory” and authored two monographs on the topic. |

PhD Supervision

PhD STUDENTS

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| 2025 – present | <p>University of Coimbra, Joint PhD Programme in Mathematics (University of Coimbra – University of Porto)</p> <ul style="list-style-type: none">• Student: Néot Choserot.• Research Project: Aspects of Two Dimensional Universal Algebra• Funding: Centre for Mathematics, University of Coimbra, Portugal. |
| 2026 – present | <p>Utrecht University, Department of Information and Computing Sciences</p> <ul style="list-style-type: none">• Student: Diogo Simm.• Research Project: Programming language semantics, Probabilistic Programming, DSLs, and Automatic Differentiation.• Co-supervised with Dr. Matthijs Vákár.• Funding: ERC Starting Grant <i>FoRECAST</i>. |
| 2026 – present | <p>University of Coimbra, Joint PhD Programme in Mathematics (University of Coimbra – University of Porto)</p> <ul style="list-style-type: none">• Student: Andrea Donati.• Research Project: A Galois Theory for Protomodular Objects• Applied for Funding |

FORMER PhD STUDENT

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| 2020 – 2024 | <p>University of Coimbra, Joint PhD Programme in Mathematics (University of Coimbra – University of Porto)</p> <ul style="list-style-type: none">• Student: Rui Rodrigues de Abreu Fernandes Prezado.• Thesis: <i>Some Aspects of Descent Theory and Applications</i>.• Research area: Generalised multicategories and Grothendieck descent theory.• Co-supervised with Prof. Maria Manuel Clementino.• Funding: FCT (Fundação para a Ciência e a Tecnologia), Portugal.• PhD awarded <i>summa cum laude</i> (January 2024). |
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Graduate Supervision

MASTER'S STUDENTS

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| 2026 – 2028 | University of Cape Town, South Africa , Master's Programme in Pure Mathematics <ul style="list-style-type: none">• Student: Kate Birch.• Research area: Category theory.• Co-supervised with Prof. Thomas Meyer. |
| 2024–2025 | IMPA, Rio de Janeiro, Brazil , Master's Programme in Pure Mathematics <ul style="list-style-type: none">• Student: Diogo Simm.• Research area: Programming language semantics, automatic differentiation, and category theory.• Co-supervised with Prof. Henrique Bursztyn and Dr. Matthijs Vákár.• Funding: National Council for Scientific and Technological Development CNPq, Brazil.• Degree awarded with distinction (Grade A). |

SHORT-TERM MASTER'S PROJECT SUPERVISION

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| 2025 | Utrecht University, The Netherlands , Scientific Methods in Computing Science (Programming Languages) <ul style="list-style-type: none">• Supervision of multiple small-group research projects at Master's level.• Topics included symbolic and automatic differentiation, program transformations, and categorical semantics.• Representative projects:<ul style="list-style-type: none">– <i>Calculating the Symbolic Derivative using Automatic Differentiation.</i>– <i>Lambda-Set Specialisation for Haskell.</i>– <i>Dual-Numbers Automatic Differentiation with Inductive Types.</i> |
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Undergraduate Supervision

UNDERGRADUATE THESIS AND CAPSTONE SUPERVISION

2023 – 2024

Utrecht University, BSc Capstone Project in Computer Science

- Students: **Cérine Toubert**, **Oscar Mensink**, **Kasper Peeters**, **Nigel Strachan**, **Joost Laan**, **Jippe Hoogeveen**, **Gust Verhees**, **Nasr Antar**, and **Artur Salek**.
- Project: Design and implementation of a software system for identifying optimal collaboration matches within organisations, based on interaction data and weighted optimisation criteria.
- Client: Dr. Ioanna Lykourantzou
- Research Areas: Optimisation algorithms, software engineering, and web-based systems.

UNDERGRADUATE SOFTWARE PROJECTS

2021–2022

Utrecht University, Game Technology

- Supervision of nine undergraduate students organised into three project teams.
- Project: Design and development of a game implemented in C#.

Teaching, Curriculum Development, and Pedagogical Leadership

COURSE DESIGN AND CURRICULUM DEVELOPMENT

2026

University of Coimbra, Portugal, BSc programme in Mathematics

- **Designed and developed the course *Object-Oriented Programming***, including the syllabus, learning outcomes, assessment structure, choice of programming language, and technical orientation.
- Structured the course as a modern introduction to object-oriented programming for mathematically trained students, emphasizing abstraction, modularity, software design, and the conceptual foundations of programming practice.
- Contributed to the revision of the *Programming Methods I and II* sequence, aligning its objectives and methodological structure with the new *Object-Oriented Programming* course and strengthening its role as a foundation for further study in programming, software development, and computational mathematics.

SELECTED TEACHING EXPERIENCE

2026 – 2027	University of Coimbra, Portugal, BSc program in Mathematics, Computer Engineering and in Artificial Intelligence and Data Science • Main Lecturer. • Course: <i>Mathematical Analysis I</i>.
2026 – 2027	University of Coimbra, Portugal, Joint PhD Programme in Mathematics (Coimbra–Porto) • Main Lecturer. • Course: <i>Categories in Algebra and Topology</i>. • Topics include enriched categories, monads, and generalised multicategories.
2026 – 2027	University of Coimbra, Portugal, BSc program in Mathematics, and Computer Engineering • Main Lecturer. • Course: <i>Object-Oriented Programming</i>.
2026 – 2027	University of Coimbra, Portugal, BSc program in Mathematics, Computer Engineering and in Artificial Intelligence and Data Science • Main Lecturer. • Course: <i>Introduction to Linear Algebra</i>.
2026 – 2027	University of Coimbra, Portugal, BSc program in Mathematics, and Computer Engineering • Main Lecturer. • Course: <i>Programming Methods I</i>.
2025 – 2026	University of Coimbra, Portugal, BSc program in Computer Engineering and in Artificial Intelligence and Data Science • Main Lecturer. • Course: <i>Mathematical Analysis II</i>.
2025 – 2026	University of Porto, Portugal, Joint PhD Programme in Mathematics (Coimbra–Porto) • Main Lecturer. • Course: <i>Categories in Algebra and Topology</i>. • Topics include enriched categories, monads, and generalised multicategories.
2020 – 2025	Utrecht University, The Netherlands, Department of Information and Computing Sciences • Graduate Teaching Assistant (Promovendus appointment). • Functional Programming — tutorials and assessment. • Logic for Computer Science — tutorials and exercises lessons. • Graphics for Computer Science (linear algebra and applications to Graphics). • Supervision of BSc Computer Science capstone software projects. • Coaching and supervision of undergraduate Game Technology projects. • Recipient of the Utrecht University Distinguished Teaching Award (2023).
2023 – 2024	Utrecht University, The Netherlands, Department of Information and Computing Sciences • Main Lecturer, BSc Computer Science. • Course: <i>Logic for Computer Science</i> (co-taught with Dr. Wouter Swierstra). • Topics included inference systems, natural deduction, semantics, and Hoare logic.

2019 – 2020	University of Porto, Portugal , Joint PhD Programme in Mathematics (Coimbra–Porto) • Main Lecturer. • Course: <i>Categories in Algebra and Topology</i>. • Topics included enriched categories, monads, and generalised multicategories.
2018 – 2019	University of Coimbra, Portugal , Department of Mathematics • Invited Assistant Professor. • Course: <i>Topology and Linear Analysis</i> (co-taught with Prof. Maria Manuel Clementino).
2017 – 2020	University of Coimbra, Portugal , Department of Mathematics • Instructor, Delfos Project. • Advanced mathematics teaching for secondary-school students preparing for national and international mathematics competitions.
2008 – 2012	University of Brasília, Brazil , Department of Mathematics • Undergraduate Teaching Assistant. • Courses included differential and integral calculus, real analysis, linear algebra, and algebra I.
2006 – 2008	Curso Dínatos, Brasília, Brazil , Secondary Education • Secondary Mathematics Teacher. • Mathematics instruction for students preparing for national university entrance examinations.
2005 – 2006	EPCAR, Brazilian Air Force, Brazil , Secondary Education • Student Teaching Assistant. • Mathematics instruction for pre-cadet students.

Invited Lectures and Talks

SELECTED INVITED AND PLENARY LECTURES

UniMont, Milan, Italy, 2026	CatAlg 2026 , Invited lecture (to be held in July 2026).
University of Messina, Italy, 2026	2nd Joint Meeting Brazil-Italy in Mathematics , Invited lecture (to be held in September 2026).
Itaca Fest, Italy 2026	Itaca Fest 2026 , Invited lecture: A Generalization of the Eilenberg–Moore Factorization.
Lorentz Center, Leiden, The Netherlands, 2022	New Challenges in Programming Language Semantics , Invited lecture: <i>A Categorical Approach to Semantic Logical Relations for Partial Features and Recursive Types</i> .
University of the Azores, Portugal, 2018	International Conference on Category Theory 2018 , Plenary lecture: <i>Aspects of Descent via Bilimits</i> .

SELECTED TALKS (INVITED AND/OR REFEREED)

- Leiden University, The Netherlands, 2024
- University of Santiago de Compostela, Spain, 2024
- University of Coimbra, Portugal, 2024
- University of Coimbra, Portugal, 2024
- University of Coimbra, Portugal, 2023
- Université catholique de Louvain, Belgium, 2023
- University of Coimbra, Portugal, 2022
- University of Primorska, Slovenia, 2021
- University of Coimbra, Portugal, 2019
- University of Edinburgh, Scotland, 2019
- Stellenbosch University, South Africa, 2019
- Masaryk University, Brno, Czech Republic, 2018
- Universidade de Santiago de Compostela, Spain, 2018
- Université catholique de Louvain, Belgium, 2017
- University of Coimbra, Portugal, 2017
- University of Coimbra, Portugal, 2016
- University of Aveiro, Portugal, 2015
- 109th Peripatetic Seminar on Sheaves and Logic**, The semantic lax descent factorization of a functor.
- International Conference in Category Theory 2024**, Doubly-infinitary distributive categories
- SUMTOPO 2024, Topology and Logic in Computer Science**, Homomorphic reverse differentiation for partial features
- 38th Summer Conference on Topology and its Applications**, Free distributive and extensive categories
- XIV Portuguese Category Seminar**, Freely generated bicartesian categories and exponentials
- International Conference in Category Theory 2023**, Spanning the tale of “Monades et descente”
- XIII Portuguese Category Seminar**, CHAD: elementary categorical aspects of automatic differentiation
- 8th European Congress of Mathematics**, Eilenberg-Moore, Kleisli and descent factorizations.
- XII Portuguese Category Seminar**, Lax comma 2-categories.
- International Conference on Category Theory 2019**, Descent and Monadicity.
- Categorical Structures in Algebra, Topology and Logic**, Eilenberg-Moore, Kleisli and descent factorization.
- 103rd Peripatetic Seminar on Sheaves and Logic**, Biadjoint Triangles and Applications.
- 102nd Peripatetic Seminar on Sheaves and Logic**, Non-canonical isomorphisms.
- 5th Workshop on Categorical Methods in Non-Abelian Algebra**, Biadjoint Triangles.
- XI Portuguese Category Seminar**, Freely generated n-categories and comonads.
- 4th Workshop on Categorical Methods in Non-Abelian Algebra**, Bilimits Commutativity and Descent
- International Conference on Category Theory 2019**, Kan extensions and descent theory.

(Invited 1-Hour Seminar) TALKS

University of Birmingham, UK, 2026	Theory of Computation Seminar , Eilenberg-Moore, Kleisli, and Descent Factorizations
IMPA, Brazil, 2025	Algebra Seminar , Biadjoint Triangles and Coherence
University of Cape Town, South Africa, 2025	Category Theory Seminar , Bifibrations, Spans, and the Bénabou-Roubaud Theorem
Stellenbosch University, South Africa, 2025	Category Theory Seminar , Two-dimensional monad theory, pseudodistributive laws, and doubly-infinitary distributive and extensive categories as pseudoalgebras
University of Cape Town, South Africa, 2025	Category Theory Seminar , Two-dimensional monad theory, and free extensive completions
University of Coimbra, Portugal, 2024	Algebra, Logic and Topology Seminar , Unraveling the iterative CHAD.
University of Coimbra, Portugal, 2022	Algebra, Logic and Topology Seminar , Descent for Split Fibrations.
Utrecht University, Netherlands, 2021	Software Technology Group Seminar , Automatic differentiation of expressive total functional languages.
University of Coimbra, Portugal, 2019	Seminar of the Category Theory Group at the University of Coimbra , On monadicity and descent.
University of Coimbra, Portugal, 2019	Algebra, Logic and Topology Seminar , Functorial Semantics and Descent II.
University of Coimbra, Portugal, 2018	Algebra, Logic and Topology Seminar , Functorial Semantics and Descent.
University of Coimbra, Portugal, 2018	Algebra, Logic and Topology Seminar , Biadjoint Triangles and Applications.
University of Coimbra, Portugal, 2018	Algebra, Logic and Topology Seminar , Towards semi left exactness and Janelidze-Galois within the lax idempotent context.
University of Coimbra, Portugal, 2016	Algebra, Logic and Topology Seminar , The deficiency of categories.
University of Coimbra, Portugal, 2015	Algebra, Logic and Topology Seminar , Biadjoint triangles, descent and coherence.
Universidade do Porto, Portugal, 2016	Research Seminar Program , Kan construction of adjunctions.
University of Coimbra, Portugal, 2016	Research Seminar Program , Homotopy Excision.
University of Coimbra, Portugal, 2013	Joint PhD Program UC-UP Seminar , Every prespectrum represents a homology theory.
Universidade de Brasília, Brazil, 2010	Math Seminar , Topological Dynamics and Applications in Number Theory (in Portuguese).
Universidade de Brasília, Brazil, 2010	Math Seminar , Dynamical Proof of the Van der Waerden Theorem (in Portuguese).

SELECTED WORKSHOP TALK

LAFI, London, UK	LAFI 2024 (Tenth Workshop on Languages for Inference at POPL, 2024) , Homomorphic Reverse Differentiation of Iteration (joint work with Matthijs Vákár and Gordon Plotkin).
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Evaluation Committees

RECRUITMENT

2025 – 2026

Hiring and Recruitment Committee, ERC Starting Grant FoRECAST

- Served as a member and advisor of the recruitment committee for three PhD positions funded by the ERC Starting Grant project *Formalised Reasoning about Expectations: Composable, Automated, Speedy, Trustworthy*.

DOCTORAL EXAMINATION COMMITTEES

2024

PhD Defense Committee, Joint University of Coimbra–University of Porto PhD Programme in Mathematics, Portugal

- Candidate: **Rui Rodrigues de Abreu Fernandes Prezado**.
- Thesis: *Some Aspects of Descent Theory and Applications*.
- Role: Supervisor and examiner.

2023

PhD Defense Committee, Joint University of Coimbra–University of Porto PhD Programme in Mathematics, Portugal

- Candidate: **Carlos Miguel Alves Fitas**.
- Thesis: *On Lax Idempotent Monads in Topology*.
- Role: Main examiner and opponent.

DOCTORAL PROJECT EVALUATION

Member of doctoral project evaluation committees within the Joint University of Coimbra–University of Porto PhD Programme in Mathematics (2019), assessing research proposals in category theory and topology, including projects on lax idempotent monads and Grothendieck descent theory. Participated as examiner and supervisor in the formal evaluation of doctoral research plans.

OTHER ACADEMIC EVALUATIONS

Evaluation of Bachelor's Theses, BSc in Artificial Intelligence, Utrecht University, The Netherlands

Academic Service and Governance

GOVERNANCE

2023 – 2025	Informatics Research Advisory Committee (IRAC) , Department of Information and Computing Sciences, Utrecht University, The Netherlands.
2020 – present	Joint University of Coimbra–University of Porto PhD Programme in Mathematics , University of Coimbra and University of Porto, Portugal.

ORGANISATION OF SCIENTIFIC EVENTS AND SEMINARS

2026 – present	Organiser, Algebra, Logic and Topology Seminar , Centre for Mathematics, University of Coimbra, Portugal.
2025 – present	Organiser, Online Algebra, Logic and Topology Seminar , Centre for Mathematics, University of Coimbra, Portugal.
2019 – 2023	Organiser, Algebra, Logic and Topology Seminar , University of Coimbra, Portugal.
2022	Organising Committee Member, XIII Portuguese Category Seminar , University of Coimbra, Portugal.
2019	Organising Committee Member, XII Portuguese Category Seminar , University of Coimbra, Portugal.
2018	Organiser, Workshop on Algebra, Logic and Topology , University of Coimbra, Portugal.
2018	Organiser, Workshop: Logic for Children , UNILOG 2018, Vichy, France.

OUTREACH AND PUBLIC ENGAGEMENT

2026 – present	CLASS Seminar , University of Coimbra, University of Cape Town, and Utrecht University I lead and organize an international seminar (called CLASS – Category, Logic, Algebra and Semantics Seminar) for graduate students and researchers working on categorical structures. The seminar brings together participants from Computer Science and Pure Mathematics, and multiple countries, including South Africa, United Kingdom, the Netherlands, Portugal, China, Brazil, and Germany, fostering collaboration and the exchange of ideas across them.
2017 – 2020	Delfos Project , Department of Mathematics, University of Coimbra, Portugal - Contributor to the Delfos Project, a mathematics education programme for secondary-school students preparing for international mathematics competitions. - Served as Portuguese representative to the International Mathematical Olympiad (IMO) in 2020.

2008 – 2012

Scientific Initiation (CNPq), Brazil

- Undergraduate member of the research group *Teoria de Lie e Dinâmica*.
- Contributed to the outreach blog <https://liedinamica.wordpress.com/>.
- Authored three educational monographs in Portuguese on Lie groups, topology for dynamical systems, and applications of topological dynamics to number theory.
- Published an expository article in *Revista Matemática Universitária* (Portuguese).

2011 – 2012

Brazilian Society for Scientific Progress (SBPC), Brazil

- Presented multiple popular-science posters in mathematics at national SBPC conferences.
- Topics included game theory, homotopy theory, category theory, topology, and dynamical systems.
- All contributions were published in the conference annals (in Portuguese).

EDITORIAL AND REFEREEING ACTIVITY

Referee for leading international journals and conferences in mathematics and computer science, including but not limited to: *ACM SIGPLAN Symposium on Principles of Programming Languages (POPL)*, *Advances in Mathematics*, *Applied Categorical Structures*, *Homology, Homotopy and Applications*, *Journal of Pure and Applied Algebra*, *Mathematical Structures in Computer Science*, *Order*, *Quaestiones Mathematicae*, *Theory and Applications of Categories*, and *Topology and its Applications*.